Chapter 4: Sequences and Series

A Level Pure Maths Notes · Rexponential

Why this chapter matters

A **sequence** is an ordered list of numbers (terms); a **series** is what you get when you add the terms of a sequence together. What's new in this chapter is the notation (particularly sigma notation), recurrence relations defined by a rule rather than a formula, and the **binomial expansion**, which needs real care, especially the negative/fractional-power version in section 4.6.

Throughout this chapter, keep two questions in mind for every problem: *is this arithmetic or geometric (or neither)?* and *am I being asked for a term, or a sum?* Mixing these up is the single most common source of lost marks.

4.1 Arithmetic sequences and series

Concept

An arithmetic sequence has a constant difference d between consecutive terms: each term is the last term plus d. This constant rate of change is exactly the discrete version of a straight line — if you plotted term number against term value, the points would lie on a line of gradient d. That's the conceptual link worth holding onto: arithmetic = linear growth.

Procedure

KEY FORMULAS

- nth term: $u_n = a + (n - 1)d$, where $a = u_1$ is the first term.
- Sum of the first n terms: $S_n = n/2 \cdot [2a + (n - 1)d] = n/2 \cdot (a + l)$, where l is the last term — the second form is just "average of first and last, times how many terms," which is quicker when you already know the last term.
- To find n when you're told a term's value, set u_n equal to that value and solve for n (it must come out as a positive integer).

Worked example

EXAMPLE 4.1

An arithmetic sequence has first term 5 and common difference 3. (a) Find the 20th term. (b) Find the sum of the first 20 terms. (c) Find the sum of all multiples of 7 between 100 and 300 (inclusive).

Step-by-step solution. (a) $u_{20} = a + 19d = 5 + 19(3) = 5 + 57 = \mathbf{62}$ (M1 for substituting into $u_n = a+(n-1)d$, A1 for 62).

(b) $S_{20} = 20/2 \cdot [2(5) + 19(3)] = 10 \cdot [10 + 57] = 10 \times 67 = \mathbf{670}$ (M1, A1).

(c) This is the type of question that trips people up because it's arithmetic *in disguise* — you first have to identify *a*, *d*, *l*, *n* yourself.

- The multiples of 7 form an arithmetic sequence with d = 7.
- First one ≥ 100: 100 ÷ 7 = 14.28…, so the first multiple is 15 × 7 = **105**. *(B1)*
- Last one ≤ 300: 300 ÷ 7 = 42.86…, so the last multiple is 42 × 7 = **294**. *(B1)*
- Number of terms: $294 = 105 + (n-1)(7) \Longrightarrow 189 = 7(n-1) \Longrightarrow n - 1 = 27 \Longrightarrow \mathbf{n = 28}$ (M1).
- $S_{28} = 28/2 \cdot (105 + 294) = 14 \times 399 = \mathbf{5586}$ (M1, A1).

EXAM TIP

Forgetting to round the endpoints *toward the inside* of the range (rounding 100÷7 down instead of up, or 300÷7 up instead of down) — always sanity-check that your first and last terms are actually multiples of *d* and actually lie inside the stated range.

4.2 Geometric sequences and series, sum to infinity

Concept

A geometric sequence has a constant *ratio* r between consecutive terms: each term is the last term times r. Where arithmetic is discrete linear growth, geometric is discrete **exponential** growth (or decay, if |r| < 1) — this is the link to compound interest, radioactive decay, and population models, and it's why geometric series matter so much in applications.

Procedure

KEY FORMULAS

- *n*th term: $u_n = ar^{n-1}$.
- Sum of the first *n* terms: $S_n = a(1 - r^n)/(1 - r)$ (equivalently $a(r^n - 1)/(r - 1)$ — use whichever avoids a negative divided by a negative).
- **Sum to infinity**: if |r| < 1, as $n \to \infty$ the term $r^n \to 0$, so the sum converges to $S_\infty = a/(1 - r)$. If |r| ≥ 1 the series diverges and S_∞ does not exist — always state the condition |r| < 1 when you use this formula; it's an easy, cheap mark that's often missed.

Worked example

EXAMPLE 4.2

(a) A geometric sequence has first term 3 and common ratio 2. Find the 10th term and the sum of the first 10 terms. (b) A geometric sequence has first term 8 and common ratio 1/4. Find the sum to infinity. (c) A geometric sequence has $u_1 = 6$ and $u_4 = 48$. Given the sequence has a positive common ratio, find r.

Step-by-step solution. (a) $u_{10} = 3 \times 2^9 = 3 \times 512 = \mathbf{1536}$. $S_{10} = 3(2^{10} - 1)/(2 - 1) = 3 \times 1023 = \mathbf{3069}$ (M1 for the formula, A1 each).

(b) Check $|r| = 1/4 < 1$, so the sum to infinity exists (B1 for stating this). $S_\infty = 8/(1 - 1/4) = 8/(3/4) = 8 \times 4/3 = \mathbf{32/3}$ (M1, A1).

(c) $u_4 = u_1 r^3$, so $48 = 6r^3 \implies r^3 = 8 \implies \mathbf{r = 2}$ (M1, A1 — remember $r^3 = 8$ only gives one real solution here since we're told r is positive; in general a cube root has one real solution anyway, unlike a square root, so no ± needed).

EXAM TIP

Using the sum-to-infinity formula without checking $|r| < 1$ — if a question gives r = 3, say, and asks for the "sum to infinity," the correct answer is that it *doesn't exist*, not a numerical value from blindly applying the formula.

4.3 Sigma notation

Concept

Sigma notation, Σ, is just compact shorthand for "add up a list of terms generated by a formula." $\sum_{r=1}^{n} f(r)$ means: substitute r = 1, 2, 3, …, n into f(r) in turn, and add up the results. It looks intimidating the first time you see it, but it's purely notation — it doesn't introduce any new mathematics, and every sigma sum in this course is either arithmetic or geometric once you expand it.

Procedure

- To **evaluate** $\sum_{r=1}^{n} f(r)$: work out the first few terms, identify whether they're arithmetic or geometric, find *a*, *d* (or *r*), and *n*, then apply the S_n formula from 4.1 or 4.2.
- To **write a sum in sigma notation**: spot the pattern in the terms (the common difference or ratio) and express the general term as a formula in *r*, then state the limits.
- Standard results worth knowing: $\sum_{r=1}^{n} 1 = n$, and more generally, splitting a sum like $\sum_{r=1}^{n} (ar + b)$ into $a\sum_{r=1}^{n} r + \sum_{r=1}^{n} b$ can simplify algebra-heavy sigma sums, though for this course direct evaluation via the AP/GP formulas is usually faster.

Worked example

EXAMPLE 4.3

(a) Evaluate $\sum_{r=1}^{6}(3r-2)$. (b) Write the sum 5 + 8 + 11 + ⋯ + 32 using sigma notation, and evaluate it.

Step-by-step solution. (a) The terms are (r=1): 1, (r=2): 4, (r=3): 7, … — an arithmetic sequence with a = 1, d = 3, n = 6 (M1 for identifying this). $S_6 = 6/2 \cdot [2(1) + 5(3)] = 3 \times 17 = \mathbf{51}$ (A1). *(Check by direct addition: 1+4+7+10+13+16 = 51 ✓.)*

(b) The terms increase by 3 each time (arithmetic), so the general term is $u_r = 5 + (r-1)(3) = 3r + 2$ (M1). The last term 32 = 3r + 2 ⟹ r = 10, so the sum is $\sum_{r=1}^{10}(3r+2)$ (A1). Evaluating: a = 5, d = 3, n = 10, $S_{10} = 10/2 \cdot (5 + 32) = 5 \times 37 = \mathbf{185}$ (M1, A1).

EXAM TIP

Misreading the lower limit — sums written with r = 1 at the bottom of ∑ versus r = 0 at the bottom give different numbers of terms and a different first term; always write out the first one or two terms explicitly to check you've matched the formula to the notation correctly.

4.4 Recurrence relations

Concept

A recurrence relation defines each term using the *previous* term(s), e.g. $u_{n+1} = f(u_n)$, rather than giving a direct formula for u_n in terms of n. This is a different way of thinking about a sequence: you generate it step by step rather than reading off any term instantly. Recurrence relations can behave in three broad ways worth recognising by eye: **increasing** (terms keep growing), **decreasing** (terms keep falling), or **periodic** (the sequence repeats itself after a fixed number of steps, called the period).

Procedure

- Given u_1 and the rule $u_{n+1} = f(u_n)$, generate terms by direct substitution: $u_2 = f(u_1)$, $u_3 = f(u_2)$, and so on.
- To classify behaviour: compute several terms and look for a pattern. If $u_{n+1} > u_n$ throughout, the sequence is increasing (and similarly for decreasing). If the values start repeating exactly, find the smallest number of steps before repetition — that's the period.
- To **prove** increasing/decreasing algebraically (rather than just observing it), consider $u_{n+1} - u_n$ and show its sign is always the same.

Worked example

EXAMPLE 4.4

(a) A sequence is defined by $u_1 = 3$, $u_{n+1} = 2u_n - 5$. Find u_2, u_3, u_4, u_5 and describe the sequence's behaviour. (b) A sequence is defined by $u_1 = 2$, $u_{n+1} = 1/(1 - u_n)$. Find u_2, u_3, u_4 and describe what you notice.

Step-by-step solution. (a) $u_2 = 2(3) - 5 = 1$. $u_3 = 2(1) - 5 = -3$. $u_4 = 2(-3) - 5 = -11$. $u_5 = 2(-11) - 5 = \mathbf{-27}$ (M1 for correct method, A1 for all values: 1, −3, −11, −27). Each term is less than the one before, so the sequence is **decreasing** (in fact it decreases without bound) (A1).

(b) $u_2 = 1/(1-2) = -1$. $u_3 = 1/(1-(-1)) = 1/2$. $u_4 = 1/(1 - 1/2) = 2 = u_1$ (M1, A1 for values −1, 1/2, 2). Since $u_4 = u_1$, the sequence repeats from here on: 2, −1, 1/2, 2, −1, 1/2, … — it is **periodic with period 3** (A1).

EXAM TIP

Assuming a sequence is periodic just because two terms happen to look similar, without checking that the *whole cycle* repeats — always confirm by generating at least one full extra period and checking every term matches.

4.5 Binomial expansion (positive integer powers)

Concept

The binomial expansion gives a formula for expanding $(a + b)^n$ without multiplying it out term by term — this is new territory, so go slowly here. The key conceptual point: the coefficients in the expansion are the entries of Pascal's triangle, and they count *how many ways* you can choose which terms contribute an *a* and which contribute a *b* when you multiply out (a+b)(a+b)⋯(a+b), n times. That combinatorial meaning is exactly what "n choose r," written ${}^nC_r$ or $\binom{n}{r}$, calculates.

Procedure

For a positive integer n:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

where $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$. In practice: write out the terms for k = 0, 1, 2, … up to n, working out $\binom{n}{k}$, the power of a (which starts at n and decreases by 1 each term), and the power of b (which starts at 0 and increases by 1 each term) — the powers of a and b always add to n in every term, which is a good self-check.

For "find the first four terms" or "find the coefficient of x^3" type questions, you only need to compute the relevant terms, not the whole expansion.

Worked example

EXAMPLE 4.5

Find the first four terms, in ascending powers of x, of the expansion of $(2 + 3x)^5$.

Step-by-step solution. Here a = 2, b = 3x, n = 5. Using (5 choose k) for k = 0,1,2,3: 1, 5, 10, 10.

- k=0: $({}^5C_0)\ 2^5\ (3x)^0 = 1 \times 32 \times 1 =$ **32**
- k=1: $({}^5C_1)\ 2^4\ (3x)^1 = 5 \times 16 \times 3x =$ **240x**
- k=2: $({}^5C_2)\ 2^3\ (3x)^2 = 10 \times 8 \times 9x^2 =$ **720x²**
- k=3: $({}^5C_3)\ 2^2\ (3x)^3 = 10 \times 4 \times 27x^3 =$ **1080x³**

(M1 for using the binomial formula with correct a, b, n; A1 for each of the four terms.)

So $(2 + 3x)^5 = 32 + 240x + 720x^2 + 1080x^3 + \cdots$ (up to the x^5 term, which we weren't asked for).

EXAM TIP

Forgetting to raise *a* to a power in every term — students often correctly track the powers of *b* (since that's "the new thing," 3x) but carelessly drop the a^{n-k} factor, or forget that a itself is being raised to a power (here 2^5, 2^4, etc.), not just 1.

4.6 Binomial expansion (negative and fractional powers)

Concept

When n is *not* a positive integer — negative, or a fraction — $(1 + x)^n$ can still be expanded as a series, but two things change fundamentally from 4.5: the expansion never terminates (it goes on forever, term after term), and it is only **valid** — meaning it actually converges to $(1+x)^n$ — for $|x| < 1$. Outside that range the series doesn't represent the function at all. This validity condition is not optional exam small print; examiners specifically test whether you state it.

Procedure

For any n (not just positive integers):

$$(1 + x)^n = 1 + nx + n(n-1)/2!\ x^2 + n(n-1)(n-2)/3!\ x^3 + \cdots \quad (|x| < 1)$$

If you're expanding $(a + bx)^n$ rather than $(1 + x)^n$, first factor out a: $(a + bx)^n = a^n(1 + (b/a)x)^n$, expand the bracket using the formula above with x replaced by (b/a)x, then multiply back through by a^n. The validity condition becomes $|(b/a)x| < 1$, i.e. $|x| < |a/b|$.

Worked example

EXAMPLE 4.6

Expand $(1 - 2x)^{-3}$ in ascending powers of x, up to and including the x^3 term, stating the range of values of x for which the expansion is valid.

Step-by-step solution. Here the expansion is directly in the form $(1 + X)^n$ with $X = -2x$ and $n = -3$ (M1 for identifying n and X correctly — this is the step where sign errors creep in).

- Constant term: 1
- x^1 term: $n \cdot X = (-3)(-2x) = 6x$
- x^2 term: $[n(n-1)/2!]\, X^2 = [(-3)(-4)/2]\,(-2x)^2 = 6 \times 4x^2 = 24x^2$
- x^3 term: $[n(n-1)(n-2)/3!]\, X^3 = [(-3)(-4)(-5)/6]\,(-2x)^3 = (-10)(-8x^3) = 80x^3$

So $(1 - 2x)^{-3} = \mathbf{1 + 6x + 24x^2 + 80x^3 + \cdots}$ *(A1 for each of the three coefficients 6x, $24x^2$ and $80x^3$ — the constant term is not separately credited)*

Validity: we need $|X| < 1$, i.e. $|-2x| < 1$, i.e. $|x| < 1/2$, so the expansion is valid for $\mathbf{-1/2 < x < 1/2}$ (M1, A1 — this mark is very commonly dropped even when the expansion itself is correct).

EXAM TIP

Losing track of signs when X itself is negative (here $X = -2x$): every odd-numbered term in the bracket $[n(n-1)(n-2)\cdots]$ picks up its own sign from n being negative, *and* every odd power of X contributes another sign — work through each factor individually rather than trying to do it in your head, and don't forget the validity condition at the end.

Chapter summary

Key formulas

KEY FORMULAS

- **Arithmetic** — $u_n = a + (n-1)d$, $S_n = n/2[2a + (n-1)d] = n/2(a + l)$
- **Geometric** — $u_n = ar^{n-1}$, $S_n = a(1 - r^n)/(1 - r)$, $S_\infty = a/(1 - r)$ $(|r| < 1)$
- **Binomial (positive integer n)** — $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$
- **Binomial (any n)** — $(1 + x)^n = 1 + nx + n(n-1)/2!\, x^2 + n(n-1)(n-2)/3!\, x^3 + \cdots$ $(|x| < 1)$

Common exam mistakes across this chapter

- Confusing arithmetic (linear) with geometric (exponential) growth — always check whether consecutive terms have a constant *difference* or a constant *ratio* before choosing a formula.
- Applying S_∞ without checking $|r| < 1$.
- In sigma notation questions, misreading the limits and getting the wrong number of terms.
- In binomial expansions with n not a positive integer, forgetting the validity condition — this is a guaranteed mark on almost every exam question of this type, so make stating it a habit, not an

afterthought.

- Sign errors in the $(1+x)^n$ expansion when x itself carries a coefficient or a minus sign — substitute carefully and work one factor at a time.

You should now be able to

- Apply the arithmetic and geometric sequence/series formulas, including sum to infinity
- Evaluate a sigma-notation sum and write a given sum in sigma notation
- Generate terms from a recurrence relation and classify its behaviour (increasing, decreasing, periodic)
- Expand $(a+b)^n$ for positive integer n, and $(1+x)^n$ for negative or fractional n, stating the validity condition